

Batch Processing Performance Analysis Report

Sales Statistics

Total Cost	Average Order	Completed	Cancelled	Processing	Pending
13,802,068.00	276.04	24864	5113	12454	7569

Executive Summary

Metric	Normal Processing	Batch Processing
Orders Processed	50000	50000
Execution Time	5.5075s	5.1782s
Peak Memory Usage	448.4081 MB	132.5013 MB
Total Memory Increase	319.8458 MB	86.3921 MB
Processing Status	Completed	Completed
Batch Count	N/A	50
Batch Size	N/A	1000

Memory Consumption Analysis

Metric	Normal Processing	Batch Processing
Start Memory (Real)	127.5564 MB	41.1633 MB
End Memory (Real)	447.4022 MB	127.5554 MB
Peak Memory (Real)	448.4081 MB	132.5013 MB
Memory Increase	319.8458 MB	86.3921 MB
Allocated Memory	458 MB	136 MB
Number of Batches	N/A	50
Average Batch Memory	N/A	1.542 MB
Largest Batch Memory	N/A	2.0284 MB
Smallest Batch Memory	N/A	1.3585 MB

Performance Analysis

Memory Reduction %: 70.31%
Batch Processing was 5.98% faster than Normal Processing.

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Performance Insights

Normal Processing loaded all orders into memory at once, causing very high memory usage of 448.4081 MB. Batch Processing kept memory stable by processing orders in chunks, reducing peak memory to 132.5013 MB. This represents a 70.31% reduction in memory usage. Batch Processing was 5.98% faster than Normal Processing. System scalability improved significantly using chunked processing, allowing handling of larger datasets without memory exhaustion.